

In-class work + Mini mock

Green Machines + Metallic materials · Year 9 Science

Each question: A / M / E

Print this sheet at 100 % (the graph grid is true size). Parts A–C after Lesson 1, Parts D–F after Lesson 2, Part G at the end of the lesson.

Part A · Flowers and seeds

1 Complete the sentences. [A]

Pollination is the transfer of pollen from the _____ to the _____. In _____ -
pollination the pollen comes from a different plant of the same species. Fertilisation happens when the
pollen _____ fuses with the ovule nucleus, forming a _____.

2 A dandelion seed has a parachute of fine hairs. Explain how this helps the dandelion plant. [M]

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3 Give one advantage and one disadvantage of a plant reproducing asexually. Explain each. [E]

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Part B · Photosynthesis and respiration

4 Complete the word equation for respiration. [A]

_____ + _____ → _____ + _____ (+ energy)

5 Before a photosynthesis experiment, a student keeps a plant in a dark cupboard for 2 days. A leaf tested with iodine then stays brown. Explain why she does this first. [M]

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6 Palisade cells are near the top of the leaf and are packed with chloroplasts. Explain how this helps the leaf. [E]

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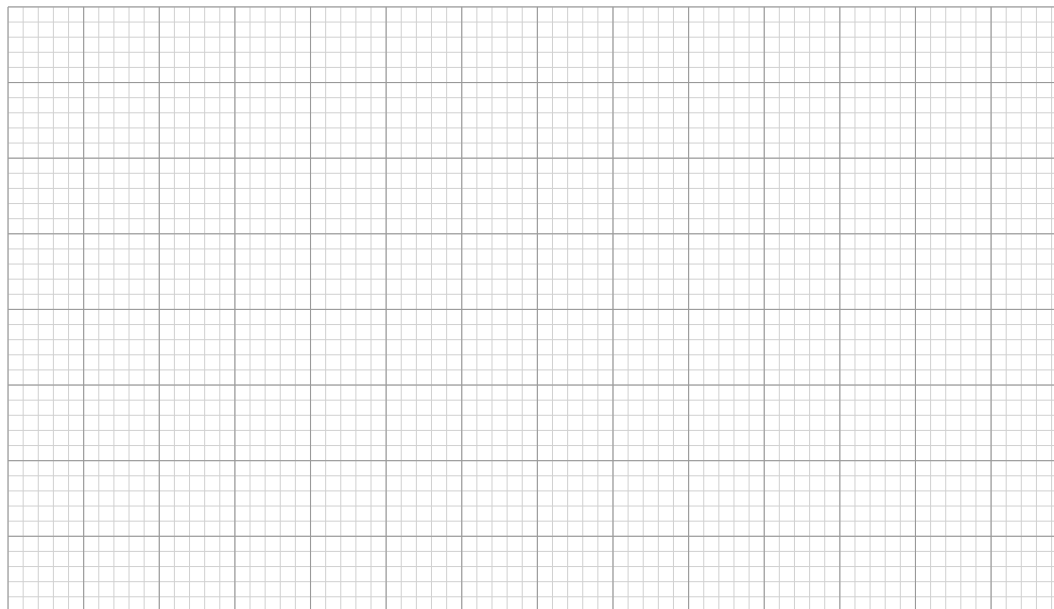
Part C · Transport and osmosis

7 The table shows the height of the liquid in a Visking tubing model.

[M]

Time (min)	0	5	10	15	20	25	30	35
Height (mm)	10	22	34	46	58	70	72	73

(a) Plot the results on the grid. Label both axes with units, choose an even scale, and draw a line.



(b) Describe the pattern in the results.

[M]

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(c) Suggest why the liquid stops rising.

[E]

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Part D · Properties and density

8 Name the property of metals that each sentence shows.

[A]

(a) Copper can be drawn out into thin wire.

(b) Gold can be beaten into very thin sheets.

(c) A brass bell rings when it is struck.

9 A metal cube has sides of 3.0 cm and a mass of 72.9 g. Calculate its density and use the booklet table (iron 7.9, copper 8.9, silver 10.5, aluminium 2.7 g/cm³) to identify the metal. Show your working. **[M]**

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Part E · Alloys and heat treatment

10 Complete: an alloy is a mixture of two or more elements, at least one of which is a **[A]**

11 Steel is iron with a small amount of carbon. Explain why steel is harder than pure iron. Use the idea of layers of atoms. **[E]**

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12 Glasses frames made of nitinol can be bent and still go back to their shape. What makes nitinol a smart alloy? **[M]**

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Part F · Corrosion and reactions

13 Write the word equations. **[A]**

- (a) calcium + hydrochloric acid →
- (b) sodium + oxygen →

14 Plan an experiment to put magnesium, iron and copper in order of reactivity, using dilute hydrochloric acid. Include your method, how you make it a fair test, what you would observe, your prediction and one safety step. **[E]**

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Part G · Mini mock (17 minutes, both units)

Answer every question. Use full sentences for every **explain** question.

15 Spelling counts in this question.

[A]

- (a) Name the tubes that carry glucose around a plant.
- (b) Name the sticky part of a flower where pollen lands.
- (c) Name the property of a metal that can be stretched into wires.

16 Explain why pea seeds need oxygen to germinate.

[M]

17 A piece of metal has a mass of 53.4 g. When it is put into a measuring cylinder, the water rises from 15.0 mL to 21.0 mL.

[A / M]

- (a) Calculate the volume of the metal.

- (b) Calculate the density of the metal. Give the unit.

18 Two leaves on the same plant are used. Leaf 1 is covered on both sides with foil. Leaf 2 is not covered. After 2 days in sunlight, both leaves are tested with iodine.

[E]

- (a) Predict the colour of each leaf after the iodine test.

- (b) Explain your predictions.

- (c) Why was Leaf 2 needed in this experiment?

19 Explain why kitchen sinks are made of stainless steel and not ordinary iron.

[M]

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20 Tin reacts slowly with dilute hydrochloric acid. Copper does not react with it.

[E]

(a) Write the word equation for tin reacting with dilute hydrochloric acid.

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(b) Copper is used for water pipes, but iron water pipes need to be protected. Explain why, using the reactivity series.

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